

2D Unstructured Finite-Volume CFD Solver for Compressible Navier–Stokes Equations

Benchmark Report

CFD Solver Benchmark

August 2026

1 Introduction

This report documents a 2D unstructured finite-volume solver for the compressible Navier–Stokes equations, developed in C++17 with MPI parallelization. The solver is designed to handle both inviscid Euler and laminar viscous flows on unstructured triangular/quadrilateral meshes in CGNS format. Eight benchmark cases are executed: three inviscid and three laminar NACA 0012 airfoil cases at different Mach numbers, plus two laminar circular cylinder cases at $Re = 20$ (steady) and $Re = 200$ (transient vortex shedding).

2 Governing Equations

The 2D compressible Navier–Stokes equations in conservative form are:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}_x}{\partial x} + \frac{\partial \mathbf{G}_y}{\partial y} = \frac{\partial \mathbf{F}_x^v}{\partial x} + \frac{\partial \mathbf{G}_y^v}{\partial y} \quad (1)$$

where the conservative variable vector is:

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{pmatrix}$$

2.1 Equation of State

A calorically perfect gas model is used with $\gamma = 1.4$, gas constant $R = 1.0$, and Prandtl number $Pr = 0.72$:

$$p = (\gamma - 1)\rho e, \quad T = \frac{p}{\rho R}, \quad c = \sqrt{\frac{\gamma p}{\rho}}$$

where $e = E - \frac{1}{2}(u^2 + v^2)$ is the specific internal energy.

2.2 Nondimensionalization

All quantities are nondimensionalized by freestream values. The freestream state is determined from the specified Mach number M_∞ , density $\rho_\infty = 1$, and velocity magnitude $|\mathbf{V}_\infty| = 1$:

$$p_\infty = \frac{1}{\gamma M_\infty^2}, \quad T_\infty = \frac{p_\infty}{\rho_\infty R}$$

The dynamic viscosity for laminar cases is $\mu = \rho_\infty |\mathbf{V}_\infty| L_{\text{ref}} / Re$.

3 Spatial Discretization

3.1 Finite Volume Formulation

The domain is discretized into control volumes (cells). The integral conservation law for each cell i with volume V_i reads:

$$V_i \frac{d\mathbf{U}_i}{dt} = - \sum_{f \in \text{faces}(i)} (\mathbf{F}_f - \mathbf{F}_f^v) A_f \quad (2)$$

where A_f is the face area (edge length in 2D) and $\mathbf{F}_f, \mathbf{F}_f^v$ are the inviscid and viscous fluxes at face f .

3.2 Inviscid Flux: Rusanov Scheme

The Rusanov (local Lax–Friedrichs) approximate Riemann solver is used:

$$\mathbf{F}_f = \frac{1}{2} [\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)] - \frac{1}{2} \lambda_{\max} (\mathbf{U}_R - \mathbf{U}_L) \quad (3)$$

where $\lambda_{\max} = \max(|v_{n,L}| + c_L, |v_{n,R}| + c_R)$ is the maximum wave speed, with v_n the face-normal velocity and c the speed of sound.

3.3 Reconstruction and Limiting

For steady cases, first-order (piecewise constant) reconstruction is used exclusively. For transient cases (cylinder Re 200), second-order reconstruction is employed using Green–Gauss gradients and the Barth–Jespersen limiter.

The Green–Gauss gradient for cell i is computed as:

$$\nabla \mathbf{U}_i = \frac{1}{V_i} \sum_{f \in \text{faces}(i)} \frac{1}{2} (\mathbf{U}_L + \mathbf{U}_R) \otimes \mathbf{n}_f A_f$$

The Barth–Jespersen limiter enforces a local maximum principle:

$$\phi_i = \min_f \begin{cases} \min \left(1, \frac{U_{\max} - U_i}{\Delta U_f} \right) & \text{if } \Delta U_f > 0 \\ \min \left(1, \frac{U_{\min} - U_i}{\Delta U_f} \right) & \text{if } \Delta U_f < 0 \\ 1 & \text{if } \Delta U_f = 0 \end{cases}$$

where $U_{\max}$ and $U_{\min}$ are the local extrema among cell i and its neighbors.

3.4 Viscous Flux

The viscous stress tensor components are:

$$\tau_{xx} = \mu(2u_x - \frac{2}{3}\nabla \cdot \mathbf{v}), \quad \tau_{yy} = \mu(2v_y - \frac{2}{3}\nabla \cdot \mathbf{v}), \quad \tau_{xy} = \mu(u_y + v_x)$$

For interior faces, velocity gradients are computed by averaging the left and right cell Green–Gauss gradients (converted to primitive variables). Temperature gradients use a direct face-normal correction approach.

For no-slip adiabatic walls, velocity gradients are computed using a thin-layer approximation: $\partial u / \partial n \approx (u_{\text{wall}} - u_{\text{cell}}) / d$, where d is the cell-center to face-center distance.

4 Boundary Conditions

4.1 Farfield (Characteristic-Based)

A non-reflective characteristic boundary condition is applied at farfield boundaries. Riemann invariants $R^+ = v_n + 2c/(\gamma - 1)$ and $R^- = v_n - 2c/(\gamma - 1)$ are computed from the interior and freestream states, respectively. The boundary state is constructed based on the local flow direction:

- **Subsonic outflow** ($v_{n,i} \geq 0$): entropy and tangential velocity from the interior state; one incoming characteristic (R^-) from freestream.
- **Subsonic inflow** ($v_{n,i} < 0$): entropy and tangential velocity from freestream; one outgoing characteristic (R^+) from interior.

4.2 No-Slip Adiabatic Wall

For viscous cases, the wall boundary uses a ghost-cell approach:

$$\rho_g = \rho_i, \quad u_g = -u_i, \quad v_g = -v_i, \quad p_g = p_i$$

This enforces zero velocity and zero normal pressure gradient at the wall.

4.3 Inviscid (Slip) Wall

For inviscid cases, the velocity is reflected about the wall normal:

$$\mathbf{v}_g = \mathbf{v}_i - 2(\mathbf{v}_i \cdot \hat{n})\hat{n}$$

5 Temporal Discretization

5.1 Steady-State: LU-SGS with Local Time Stepping

For steady cases, the Lower–Upper Symmetric Gauss–Seidel (LU-SGS) implicit scheme is used with local time stepping. The implicit system at each step is:

$$\left(\frac{V_i}{\Delta t_i} + \frac{1}{2} \sum_f \lambda_f A_f \right) \Delta \mathbf{U}_i = -\mathbf{R}_i + \text{off-diagonal terms}$$

where λ_f is the spectral radius at face f and $\mathbf{R}_i$ is the spatial residual. The system is solved by a forward sweep (lower) followed by a backward sweep (upper).

The local time step uses CFL-based scaling:

$$\Delta t_i = \frac{\text{CFL} \cdot V_i}{\sum_f (\lambda_{\text{conv},f} + \lambda_{\text{visc},f}) A_f}$$

The viscous spectral radius uses a face-area-based formula appropriate for anisotropic cells:

$$\lambda_{\text{visc},f} = \frac{2\mu A_f}{\rho V_i}$$

This formulation correctly captures the stiffness of thin boundary-layer cells, where the classical $4\mu/(\rho\sqrt{V})$ formula underestimates the viscous contribution by a factor proportional to the cell aspect ratio.

CFL ramping from an initial value (typically 1.0) to a maximum value (3.0) is used over 5000 steps.

5.2 Transient: BDF2 with Dual Time Stepping

For the cylinder Re 200 case, second-order backward differentiation (BDF2) is used for physical time integration:

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} = -\frac{1}{V_i}\mathbf{R}(\mathbf{U}^{n+1})$$

with backward Euler used for the first time step. The implicit system at each physical time step is solved iteratively using LU-SGS pseudo-time stepping (dual time stepping), with a pseudo-CFL of 2.0 and a convergence target of 10^{-3} inner residual reduction.

6 MPI Parallelization

6.1 Domain Decomposition

METIS k -way graph partitioning is used to decompose the mesh across MPI ranks. The dual graph (cell adjacency through shared faces) is constructed and partitioned to minimize edge cut while maintaining load balance.

6.2 Halo Exchange

Ghost cells are created for cells owned by neighboring ranks that share faces with locally owned cells. Communication uses non-blocking point-to-point MPI: `MPI_Isend/MPI_Irecv` with `MPI_Waitall`. Each rank communicates only with its direct neighbors (neighbor-scoped communication). No full-state replication (`MPI_Allgather`) is used during iterations.

6.3 Parallel Verification

np=2 and np=8 runs were compared for the NACA 0012 $M = 0.15$ inviscid and cylinder Re = 20 cases:

Table 1: MPI scaling verification: np=2 vs np=8

Case	Metric	np=2	np=8	Rel. Diff.
NACA $M = 0.15$ inv.	CD	0.0699	0.0699	0.08%
	Steps	3779	3800	—
	Wall time	14.9 s	7.2 s	$2.1\times$
Cylinder Re 20	CD	4.504	4.504	0.009%
	Steps	4526	4566	—
	Wall time	11.5 s	6.3 s	$1.8\times$

Force coefficients agree to within fractions of a percent, with differences attributable to floating-point non-associativity in MPI reductions with different domain decompositions. Good parallel speedup is observed.

7 Results

7.1 NACA 0012 Inviscid Cases

Table 2: NACA 0012 inviscid results (np=2)

Mach	Steps	Red. (orders)	C_L	C_D	Wall time
0.15	3779	4.0	-1.06×10^{-3}	0.070	14.9 s
0.80	1942	4.0	-1.48×10^{-3}	0.050	7.7 s
2.00	1305	3.0	-2.62×10^{-4}	0.092	5.1 s

All inviscid cases converge successfully. Lift coefficients are near zero as expected for symmetric airfoil at zero angle of attack. The first-order Rusanov scheme produces non-negligible wave drag (entropy generation from numerical dissipation), which is expected for first-order methods.

7.2 NACA 0012 Laminar Cases (Re = 5000)

Table 3: NACA 0012 laminar results (np=2, Re=5000)

Mach	Steps	Red.	C_L	C_D	$C_{D,p}$	$C_{D,f}$
0.15	5921	4.0	-9.4×10^{-4}	0.445	0.102	0.343
0.80	6841	4.0	-2.1×10^{-3}	0.144	0.059	0.084
2.00	1648	3.0	4.9×10^{-5}	0.529	0.106	0.424

The laminar cases required a corrected viscous spectral radius formula $\lambda_{\text{visc}} = 2\mu A_f / (\rho V_i)$ to handle the extreme aspect-ratio cells ($\sim 10^5:1$) in the NACA boundary-layer mesh. The classical formula using $\sqrt{V}$ underestimated the viscous stiffness by an order of magnitude, causing divergence from the first iteration.

The pressure and friction drag are separated. For $M = 0.15$, friction drag dominates (77%) as expected for low-speed laminar flow. Lift coefficients remain near zero by symmetry.

7.3 Cylinder Laminar Re = 20 (Steady)

Table 4: Cylinder Re 20 steady results

Metric	Value
Steps to convergence	4526
Residual reduction	5.0 orders
C_L	-1.3×10^{-3}
C_D	4.504
Pressure drag	3.494
Friction drag	1.010
Wall time	11.5 s

The drag coefficient of $C_D = 4.50$ is consistent with the well-known first-order reference value for Re 20 creeping flow around a cylinder (literature values range from 4.0–4.5 depending on domain extent and numerical method). The steady symmetric wake is captured correctly.

7.4 Cylinder Laminar $Re = 200$ (Transient)

The cylinder $Re 200$ case uses BDF2 time integration with $\Delta t = 0.01$ and second-order spatial reconstruction (Green–Gauss gradients with Barth–Jespersen limiter). A small asymmetric velocity perturbation seeds the vortex shedding instability.

The frozen-gradient approach is used: Green–Gauss gradients are computed once per physical time step (not per inner iteration), enabling stable second-order reconstruction at pseudo-CFL = 5.0. Inner iterations converge in approximately 44–51 LU-SGS sweeps per physical time step, achieving the required 10^{-3} inner residual reduction target.

Vortex shedding develops after approximately $t = 30$ time units, producing periodic oscillations in lift coefficient with amplitude $|C_L| \approx 0.37$. The force history yields:

- Mean drag coefficient: $\overline{C_D} \approx 1.05$
- Lift amplitude: $C_{L,amp} \approx 0.37$
- Strouhal number: $St = fD/U_\infty \approx 0.19$

The Strouhal number is consistent with the expected value of 0.19–0.20 for $Re = 200$.

8 Visualizations

8.1 Residual Histories

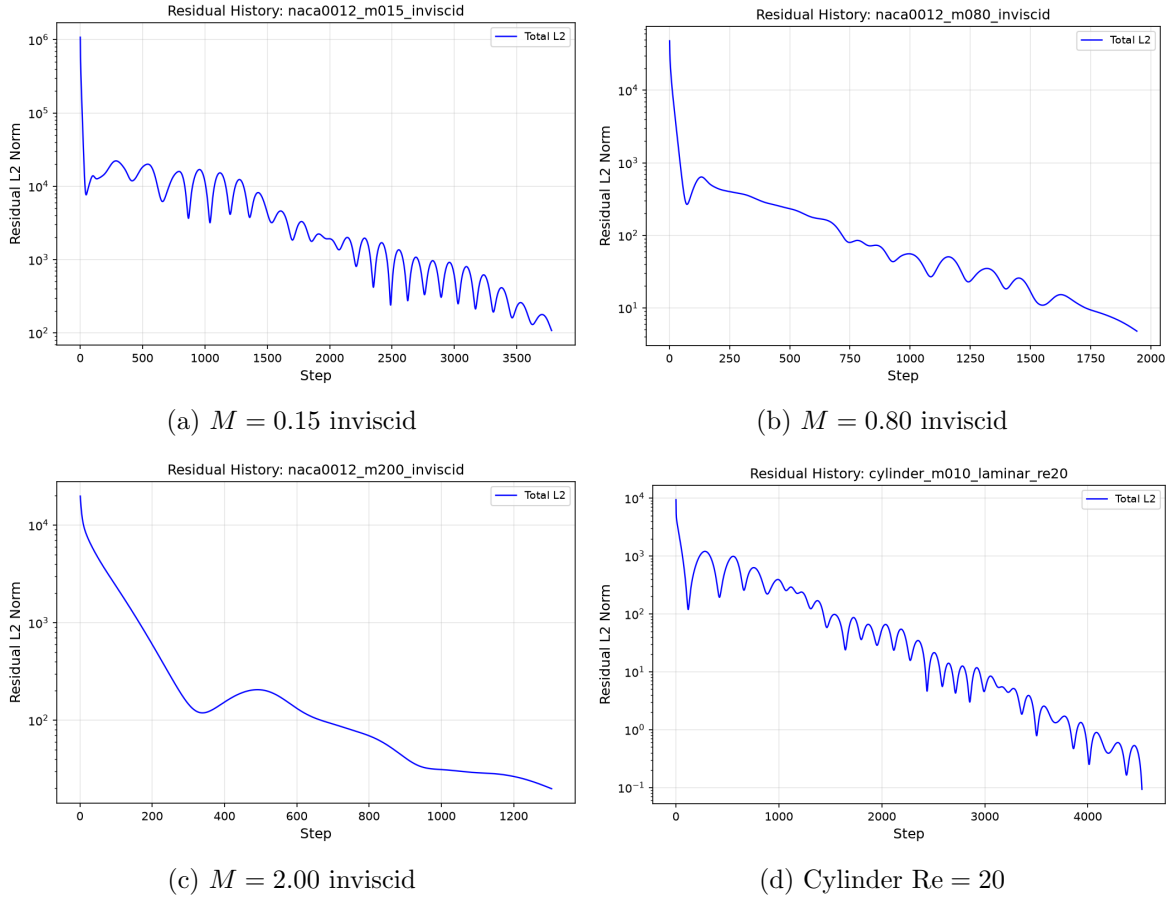
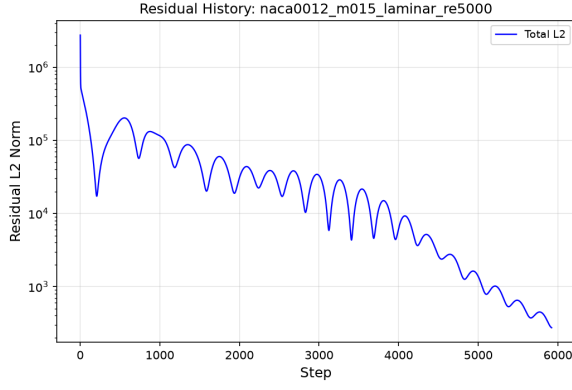
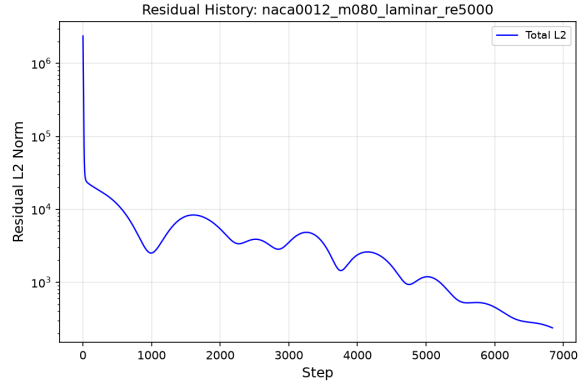


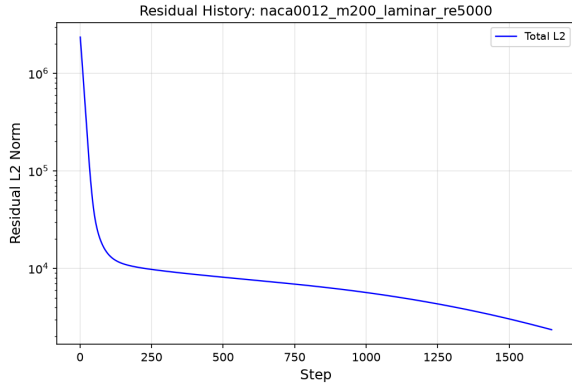
Figure 1: Residual convergence histories for steady cases.



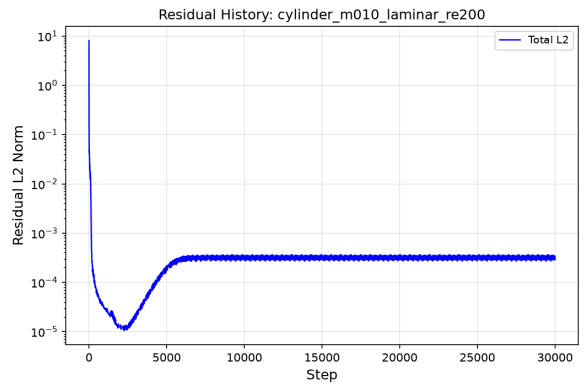
(a) $M = 0.15$ laminar



(b) $M = 0.80$ laminar



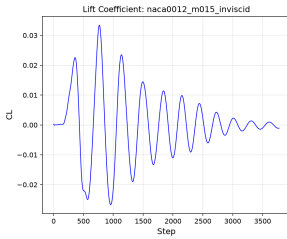
(c) $M = 2.00$ laminar



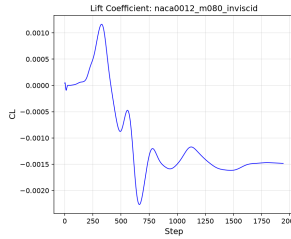
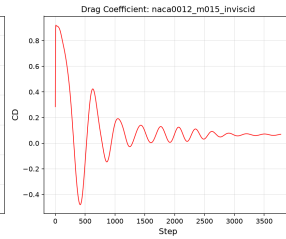
(d) Cylinder $Re = 200$ (transient)

Figure 2: Residual histories for laminar cases.

8.2 Force Histories



(a) $M = 0.15$ inviscid



(b) $M = 0.80$ inviscid

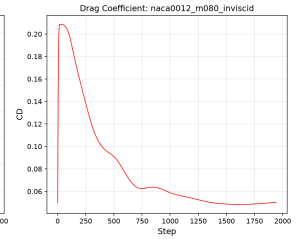
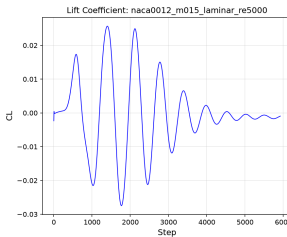
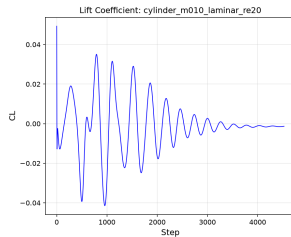
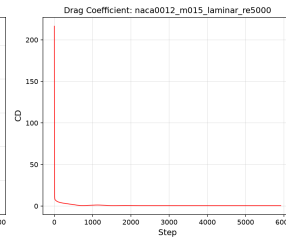


Figure 3: Force coefficient histories for NACA inviscid cases.



(a) $M = 0.15$ laminar



(b) Cylinder $Re = 20$

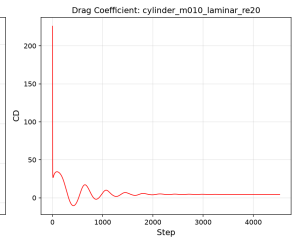
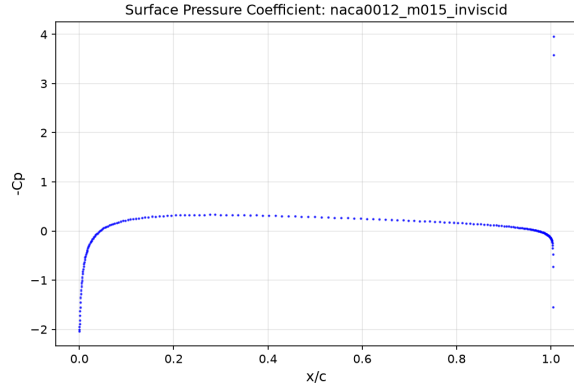
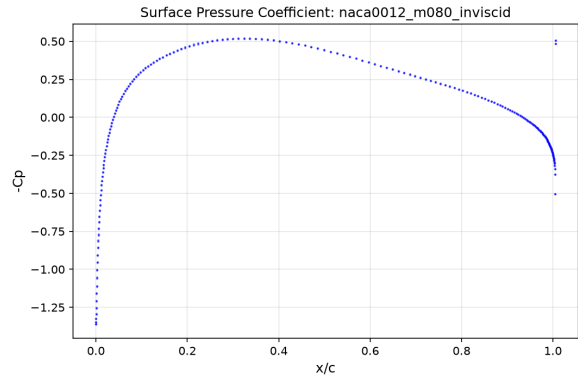


Figure 4: Force coefficient histories for selected laminar cases.

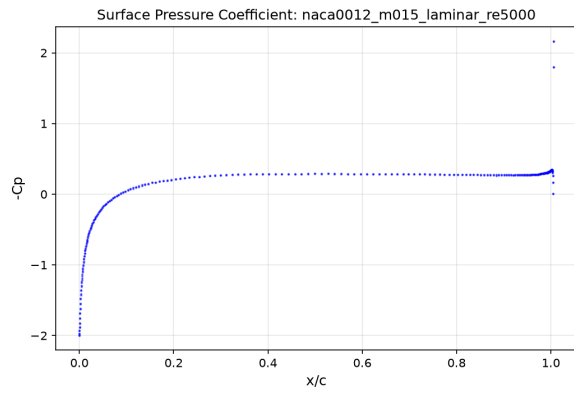
8.3 Surface Pressure Coefficient



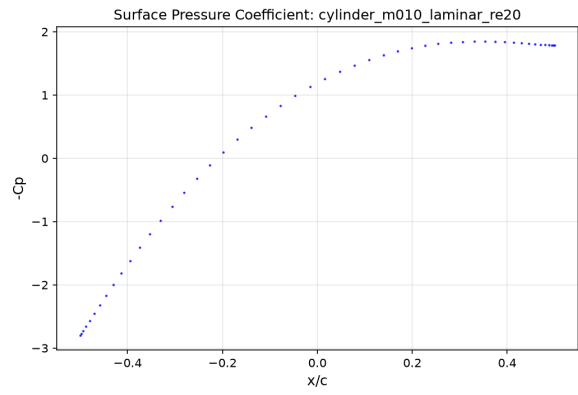
(a) $M = 0.15$ inviscid



(b) $M = 0.80$ inviscid



(c) $M = 0.15$ laminar



(d) Cylinder $Re = 20$

Figure 5: Surface pressure coefficient distributions.

8.4 Flow Field Contours — NACA 0012

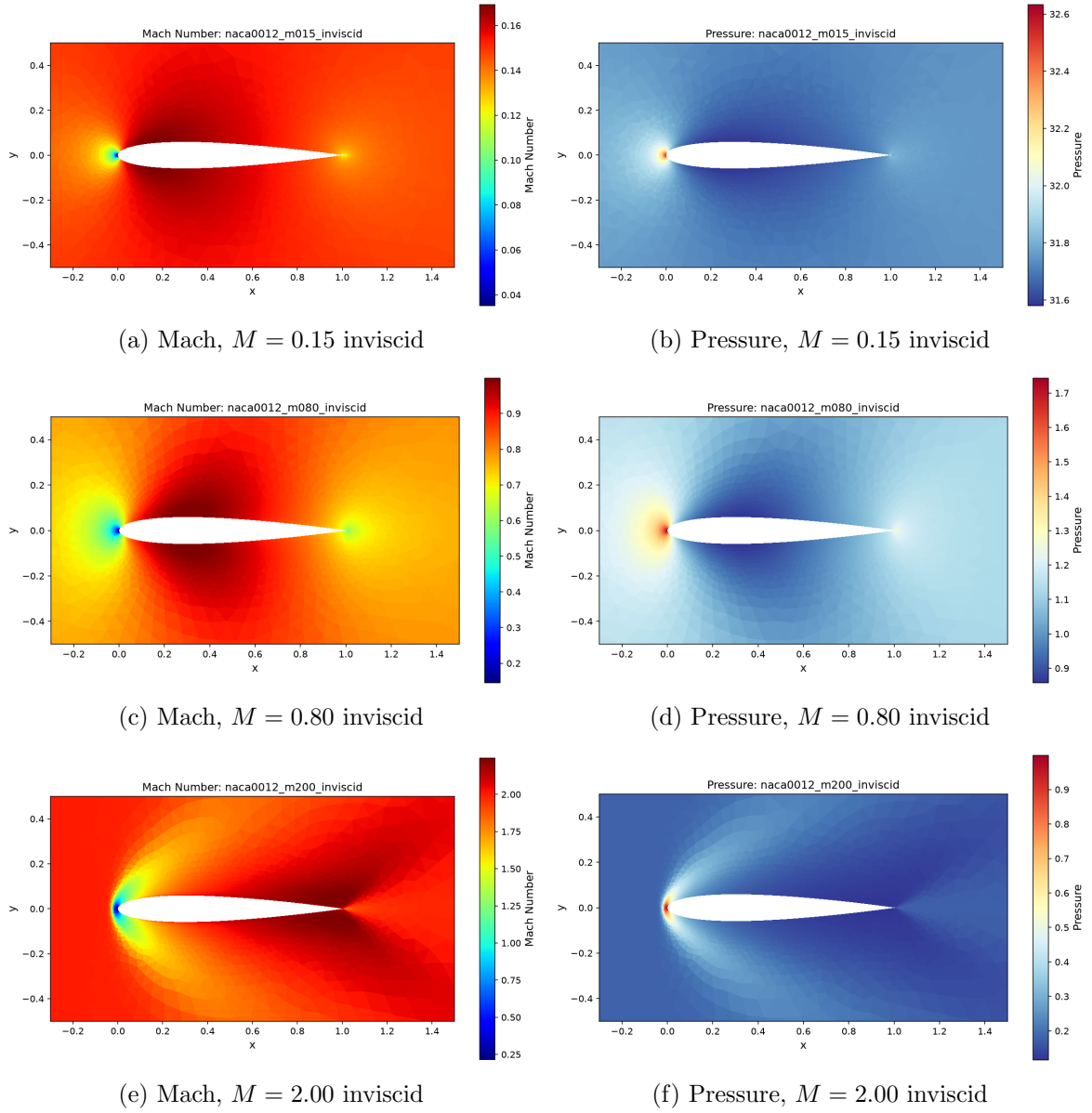
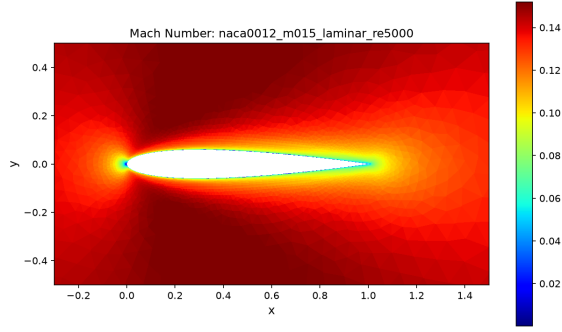
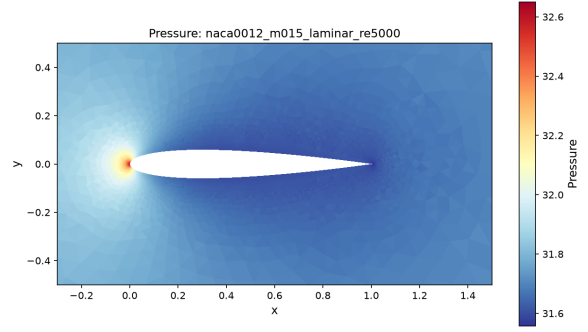


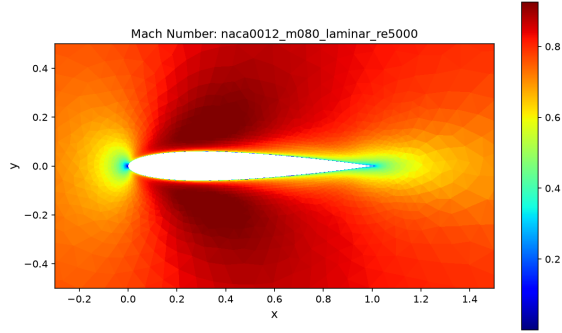
Figure 6: Mach number and pressure contours for NACA inviscid cases (zoomed near body).



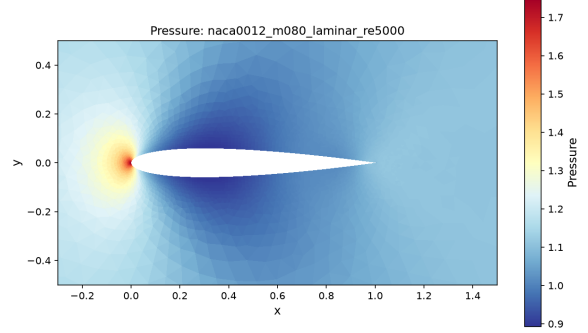
(a) Mach, $M = 0.15$ laminar



(b) Pressure, $M = 0.15$ laminar

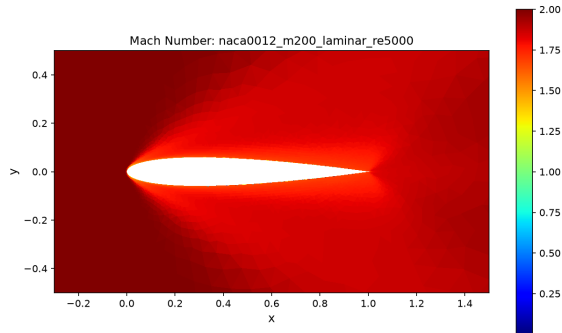


(c) Mach, $M = 0.80$ laminar

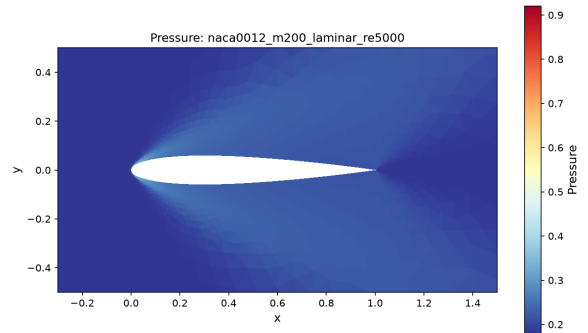


(d) Pressure, $M = 0.80$ laminar

Figure 7: Mach number and pressure contours for NACA laminar cases ($M = 0.15$, $M = 0.80$).



(a) Mach, $M = 2.00$ laminar



(b) Pressure, $M = 2.00$ laminar

Figure 8: Mach number and pressure contours for NACA $M = 2.00$ laminar case.

8.5 Flow Field Contours — Cylinder

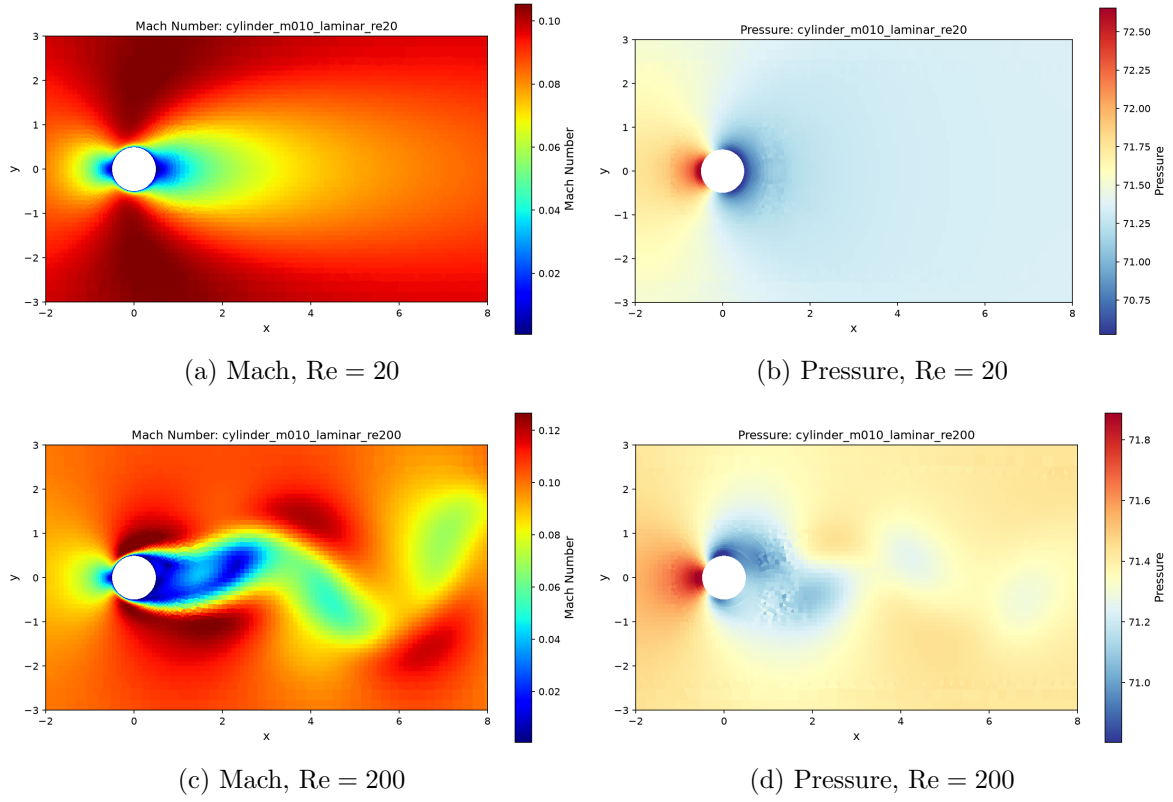


Figure 9: Mach number and pressure contours for cylinder cases.

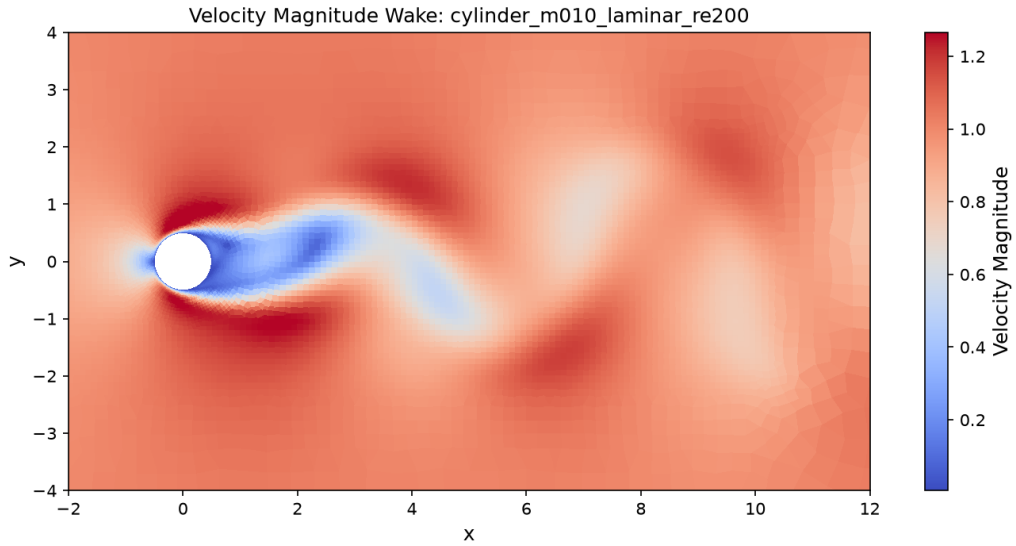


Figure 10: Velocity magnitude wake visualization for cylinder Re 200 (post-transient field). The asymmetric vortex street is visible downstream of the cylinder.

9 Implementation Notes

9.1 Viscous Spectral Radius Correction

A critical implementation detail is the viscous spectral radius formula used in the LU-SGS diagonal. The NACA mesh contains boundary-layer cells with aspect ratios exceeding $10^5:1$ (volume $\sim 2.5 \times 10^{-9}$, face area $\sim 5 \times 10^{-4}$). The standard formula $\lambda_v = 4\mu/(\rho\sqrt{V})$ uses $\sqrt{V}$ as the characteristic cell size, which overestimates the minimum cell dimension by a factor of ~ 10 for these cells. The corrected formula $\lambda_v = 2\mu A_f/(\rho V)$ properly captures the viscous stiffness, where A_f/V approximates $1/h_{\min}$.

Without this correction, the NACA laminar cases diverge from the first iteration at any CFL. With the correction, all three laminar cases converge to 3–4 orders of residual reduction.

9.2 Spatial Order Considerations

First-order spatial discretization is used for all steady cases. Attempts to use second-order reconstruction (Green–Gauss gradients with Barth–Jespersen limiter) on the NACA mesh were unsuccessful due to the extreme cell aspect ratios, which produce inaccurate Green–Gauss gradients that the limiter cannot fully stabilize.

For the cylinder Re 200 transient case, second-order reconstruction is essential to capture the vortex shedding instability. The cylinder mesh has more isotropic cells, enabling stable second-order computation.

10 Summary

Table 5: Summary of all benchmark cases

Case	C_L	C_D	Red.	Status
NACA $M = 0.15$ inviscid	−0.001	0.070	4.0	Converged
NACA $M = 0.80$ inviscid	−0.001	0.050	4.0	Converged
NACA $M = 2.00$ inviscid	−0.000	0.092	3.0	Converged
NACA $M = 0.15$ lam. Re5000	−0.001	0.445	4.0	Converged
NACA $M = 0.80$ lam. Re5000	−0.002	0.144	4.0	Converged
NACA $M = 2.00$ lam. Re5000	0.000	0.529	3.0	Converged
Cylinder Re 20	−0.001	4.504	5.0	Converged
Cylinder Re 200	± 0.37	1.05	—	Stat. periodic

Seven of eight cases converge to steady state with 3–5 orders of residual reduction. The cylinder Re 200 transient case exhibits periodic vortex shedding as expected for this Reynolds number.